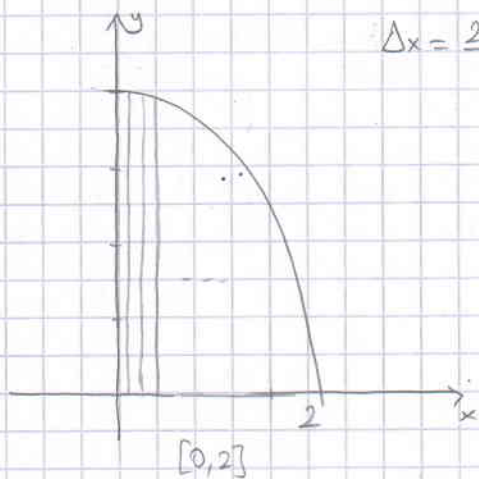


DEFINITE INTEGRAL

① Find the area of the region in the 1st quadrant bounded by $f(x) = 4 - x^2$ and the lines $x = 0$ and $y = 0$



$$\Delta x = \frac{2-0}{n} = \frac{2}{n} \quad \sum_{i=1}^n f(x_i) \Delta x = f(x_1) \Delta x + f(x_2) \Delta x + \dots + f(x_n) \Delta x$$

$$= f\left(\frac{2}{n}\right) \frac{2}{n} + f\left(\frac{4}{n}\right) \frac{2}{n} + \dots + f\left(\frac{2n}{n}\right) \frac{2}{n}$$

$$= \frac{2}{n} \left[4 - \left(\frac{2}{n}\right)^2 + 4 - \left(\frac{4}{n}\right)^2 + \dots + 4 - \left(\frac{2n}{n}\right)^2 \right]$$

$$= \frac{2}{n} \cdot 4 \cdot \left[1 - \left(\frac{1}{n}\right)^2 + 1 - \left(\frac{2}{n}\right)^2 + \dots + 1 - \left(\frac{n}{n}\right)^2 \right]$$

$$= \frac{8}{n} \left[n - \frac{1}{n^2} (1^2 + 2^2 + \dots + n^2) \right]$$

$$= \frac{8}{n} \left[n - \frac{1}{n^2} \cdot \frac{n(n+1)(2n+1)}{6} \right]$$

$$\sum_{i=1}^n f(x_i) \Delta x = 8 - \frac{4(n+1)(2n+1)}{3n^2} = 8 - \frac{8n^2 + 3n + 1}{3n^2}$$

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x = \lim_{n \rightarrow \infty} \left(8 - \frac{8n^2 + 3n + 1}{3n^2} \right) = 8 - \frac{8}{3} = \frac{16}{3} \text{ br}^2$$

$$\int_0^2 (4 - x^2) dx = \left[4x - \frac{x^3}{3} \right]_0^2 = 8 - \frac{8}{3} = \frac{16}{3} \text{ br}^2$$

② Determine the following integral $\int_0^3 (9 - x^2) dx$ using the Riemann sum

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x \quad \Delta x = \frac{b-a}{n} \quad x_i = a + i \Delta x, \quad i = 0, 1, 2, \dots, n$$

$$\Delta x = \frac{3-0}{n} = \frac{3}{n} \quad x_i = 0 + i \frac{3}{n}, \quad i = 0, 1, 2, \dots, n$$

$$\int_0^3 (9 - x^2) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f\left(\frac{3i}{n}\right) \cdot \frac{3}{n} = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(9 - \frac{9i^2}{n^2} \right) \cdot \frac{3}{n} = \lim_{n \rightarrow \infty} \left\{ 27 - \frac{27}{n^3} \sum_{i=1}^n i^2 \right\}$$

$$= \lim_{n \rightarrow \infty} \left\{ 27 - \frac{27}{n^3} \sum_{i=1}^n i^2 \right\} = \lim_{n \rightarrow \infty} \left\{ 27 - \frac{27}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} \right\}$$

$$= \lim_{n \rightarrow \infty} \left\{ 27 - \frac{9(n+1)(2n+1)}{2n^2} \right\} = 27 - \frac{18}{2} = 27 - 9 = 18 \text{ br}^2$$

③ Evaluate $\int_0^3 (x-5)dx$ by using the definition of definite integral

$$\Delta x = \frac{3-0}{n} = \frac{3}{n} \quad x_i = a + i\Delta x = 0 + i \cdot \frac{3}{n} = \frac{3i}{n}$$

$$\begin{aligned} \int_0^3 (x-5)dx &= \lim_{n \rightarrow \infty} \sum_{i=1}^n f\left(\frac{3i}{n}\right) \cdot \frac{3}{n} = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{3i}{n} - 5\right) \cdot \frac{3}{n} \\ &= \lim_{n \rightarrow \infty} \left\{ \frac{9}{n^2} \sum_{i=1}^n i - \frac{15}{n} \sum_{i=1}^n 1 \right\} = \lim_{n \rightarrow \infty} \left\{ \frac{9}{n^2} \cdot \frac{n(n+1)}{2} - \frac{15}{n} \cdot n \right\} \\ &= \lim_{n \rightarrow \infty} \left(\frac{9n+9}{2} - 15 \right) = \frac{9}{2} - 15 = -\frac{21}{2} \end{aligned}$$

④ **Theorem:** Suppose that the function f is continuous on an interval I containing the point a .

Part-1: Let the function F be defined on I by $F(x) = \int_a^x f(t)dt$

Then F is differentiable on I and $F'(x) = f(x)$ there. Thus F is antiderivative of f on I :

$$\frac{d}{dx} \int_a^x f(t)dt = f(x)$$

Part-2: If $G(x)$ is any antiderivative of $f(x)$ on I , so that $G'(x) = f(x)$ on I , then for any b in I we have

$$\int_a^b f(x) = G(b) - G(a)$$

Proof-1: Using the definition of the derivative

$$\begin{aligned} F'(x) &= \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \left\{ \int_a^{x+h} f(t)dt - \int_a^x f(t)dt \right\} = \lim_{h \rightarrow 0} \frac{1}{h} \left\{ \int_x^{x+h} f(t)dt \right\} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \cdot h \cdot f(c) = \lim_{h \rightarrow 0} f(c) = \lim_{c \rightarrow x} f(c) = f(x) \end{aligned}$$

$$\begin{array}{ccc} x & c & x+h \\ \text{while } h \rightarrow 0, & c \rightarrow x & \end{array}$$

Proof-2: $\int_a^x f(t)dt = F(x) = G(x) + c$

$$x=a \text{ then } 0 = G(a) + c \Rightarrow c = -G(a)$$

$$x=b \text{ then } \int_a^b f(t)dt = G(b) + c = G(b) - G(a)$$

$$\textcircled{5} \int_{\sqrt{e/3}}^{\sqrt{4e/3}} x e^{x^2+1} dx = \frac{1}{2} \int e^u du = \frac{1}{2} e^u \Big|_{\sqrt{e/3}}^{\sqrt{4e/3}} = \frac{1}{2} [e^{4u+1} - e^{u+1}]$$

$$= \frac{1}{2} [4e - 3e]$$

$$x^2+1=u$$

$$2x dx = du$$

$$\text{veya } u_1 = (\sqrt{e/3})^2 + 1 = \ln(3e), \quad u_2 = (\sqrt{4e/3})^2 + 1 = \ln(4e)$$

$$\frac{1}{2} \int_{\ln 3e}^{\ln 4e} e^u du = \frac{1}{2} e^u \Big|_{\ln 3e}^{\ln 4e} = \frac{e^{\ln 4e} - e^{\ln 3e}}{2} = \frac{4e - 3e}{2} = \frac{e}{2}$$

$$\textcircled{6} \int_e^{e^2} \frac{dx}{x \ln x} = \ln(\ln x) \Big|_e^{e^2} = \ln(\ln e^2) - \ln(\ln e) = \ln 2 - \ln 1 = \ln 2$$

$\textcircled{7}$ Find the average value of the function $f(x) = \frac{1}{x}$ on the interval $[1, e]$

$$\bar{f} = \frac{1}{b-a} \int_a^b f(x) dx$$

$$\bar{f} = \frac{1}{e-1} \int_1^e \frac{1}{x} dx = \frac{1}{e-1} \ln x \Big|_1^e = \frac{\ln e - \ln 1}{e-1} = \frac{1}{e-1}$$

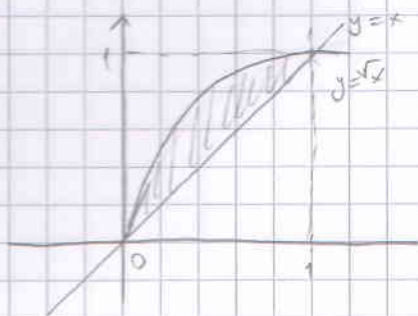
$$\textcircled{8} f(x) = \begin{cases} x^{4/3} & x < 0 \\ \sqrt{x}(x+1) & x \geq 0 \end{cases} \quad I = \int_{-1}^1 f(x) dx = ?$$

$$I = \int_{-1}^1 f(x) dx = \int_{-1}^0 x^{4/3} dx + \int_0^1 \sqrt{x}(x+1) dx = \frac{x^{1/3+1}}{1/3+1} \Big|_{-1}^0 + \int_0^1 (x^{3/2} + x^{1/2}) dx$$

$$= \frac{3}{4} x^{4/3} \Big|_{-1}^0 + \left(\frac{2}{5} x^{5/2} + \frac{2}{3} x^{3/2} \right) \Big|_0^1 = \left(0 - \frac{3}{4} \right) + \left(\frac{2}{5} + \frac{2}{3} \right) = \frac{40+24-45}{60} = \frac{19}{60}$$

AREA BETWEEN CURVES

- ① Find the area of the region bounded by the curves $y = \sqrt{x}$ and $y = x$



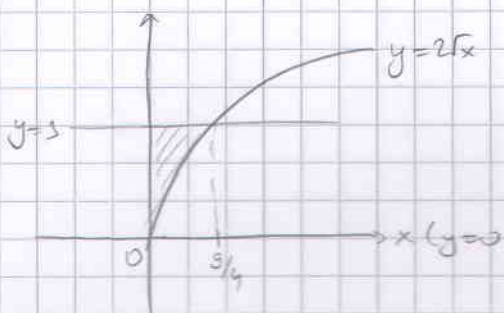
$$\sqrt{x} = x \Rightarrow x = x^2 \Rightarrow x^1 - x = 0 \quad x(x-1) = 0$$

$$x_1 = 0 \quad x_2 = 1$$

$$A = \int_0^1 (\sqrt{x} - x) dx = \left(\frac{2}{3} x^{3/2} - \frac{x^2}{2} \right) \Big|_0^1$$

$$= \frac{2}{3} - \frac{1}{2} = \frac{1}{6} \text{ br}^2 //$$

- ② Find the area of the region bounded by the curve $y^2 = 4x$ and the line $y = 3$ and $x = 0$



$$\text{I. Yol.} \int_0^{3/4} (3 - 2\sqrt{x}) dx = \left(3x - \frac{4}{3} x^{3/2} \right) \Big|_0^{9/4}$$

$$= \frac{27}{4} - \frac{4}{3} \cdot \frac{27}{8} = \frac{9}{4} \text{ br}^2 //$$

$$\text{II. Yol.} \int_0^3 \left(\frac{y^2}{4} - 0 \right) dy = \frac{y^3}{12} \Big|_0^3 = \frac{27}{12} = \frac{9}{4} \text{ br}^2 //$$